

1 The Classical situation

Imagine two beams of entangled photons going into two detectors. One detector, A, is at angle θ and the other, B, is at angle ϕ to the horizontal. Modelling the photons as polarised either horizontally or vertically. Malus' law states that the probability of detector A measuring polarisation along the X axis is the square of the projection of the polarisation onto the direction of A.

$$P(X_A) = \cos^2 \theta$$

A is aligned $90 - \theta$ to the vertical so:

$$P(Y_A) = \cos^2(90 - \theta) = \sin^2 \theta$$

A useful observable is the probability of detecting matching polarisations:

$$P(X_A)P(X_B) + P(Y_A)P(Y_B) = \cos^2 \theta \cos^2 \phi + \sin^2 \theta \sin^2 \phi$$

and of detecting mismatching ones:

$$1 - \cos^2 \theta \cos^2 \phi - \sin^2 \theta \sin^2 \phi$$

2 The Quantum one

One element of the copenhagen interpretation of quantum mechanics is that the wavefunction can be written as a sum of the eigenstates corresponding to all possible outcomes of a measurement and 'collapses' into one of them after a measurement occurs. This is interesting since the schrodinger equation does not seem to describe any collapse. For our purposes, however, the copenhagen interpretation implies that measuring one photon's polarisation and then the others is not independent since in between the two photon state collapses due to the first measurement. Thus:

$$P(X_A, X_B) = \cos^2 \theta \cos^2(\phi - \theta)$$

where we have replaced what would be $\cos \phi$ in the classical case with $\cos(\phi - \theta)$ since the wavefunction has collapsed so that polarisation is along the X_A direction, which is an angle $\phi - \theta$ from X_B . Noting that Y_A is also $\phi - \theta$ from Y_B we have:

$$P(Y_A, Y_B) = \sin^2 \theta \cos^2(\phi - \theta)$$

But now the probability of observing matching polarisations is simply:

$$P(X_A, X_B) + P(Y_A, Y_B) = \cos^2(\phi - \theta)(\cos^2 \theta + \sin^2 \theta) = \cos^2(\phi - \theta)$$

Making the probability of mismatching ones:

$$1 - \cos^2(\phi - \theta)$$

You may have worried that we chose for it to be measured by detector A first, even though we want to do a calculation for measurement by A and B simultaneously. Since $\cos(-x) = \cos(x)$, the probability of matching (and thus also mismatching) polarisations is the same as if we swap ϕ and θ meaning we could have assumed it was measured by B first and the probability would come out the same.